Negative results and claim revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Robustness across conditions. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of negative results and claim revision. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **negative result**s and **claim revision**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **negative result**s and **claim revision**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Robustness across conditions. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **negative result**s and **claim revision** using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Negative results and claim revision case. Separate observations, given facts and assumptions before choosing an explanation.

Negative results report that the **evidence** did not meet a declared expectation. They can reveal design limits or useful constraints without proving no effect anywhere.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Negative results and claim revision. Connect each labelled input, deciding step and result.

Compare results with the original criterion and uncertainty assumptions. Preserve the failed hypothesis, discuss plausible **mechanism**s and label revised questions as new work.

Negative results and claim revision — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Hypothesis expects B support strictly above A. Both pass8/10 reserved cases.	Compare results with the original criterion and uncertainty assumptions. Preserve the failed hypothesis, discuss plausible mechanisms and label revised questions as new work.	Strict improvement is not observed; both80%. This does not prove population equality or universal uselessness.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Negative results and claim revision. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Hypothesis expects B support strictly above A. Both pass8/10 reserved cases. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: B wins on one new task after the team chooses it because A fails.

Predicted result

Prediction: Treat that selected demonstration as exploratory, not a replacement for the original result.

Why the result follows

Reason: Case selection now depends on outcomes.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Negative results and claim revision to explain a concrete decision,

> rather than only memorising definitions.

Word	Meaning
Negative result	Evidence failing to meet the declared expectation
Claim revision	A bounded change to the conclusion
Equivalence	A separate claim requiring its own design and criterion

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Negative results and claim revision, predict a step, then compare with its explanation.

Hypothesis expects B support strictly above A. Both pass8/10 reserved cases.

Strict improvement is not observed; both**80%**. This does not prove population equality or universal uselessness.

Worked explanation · Negative results and claim revision

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

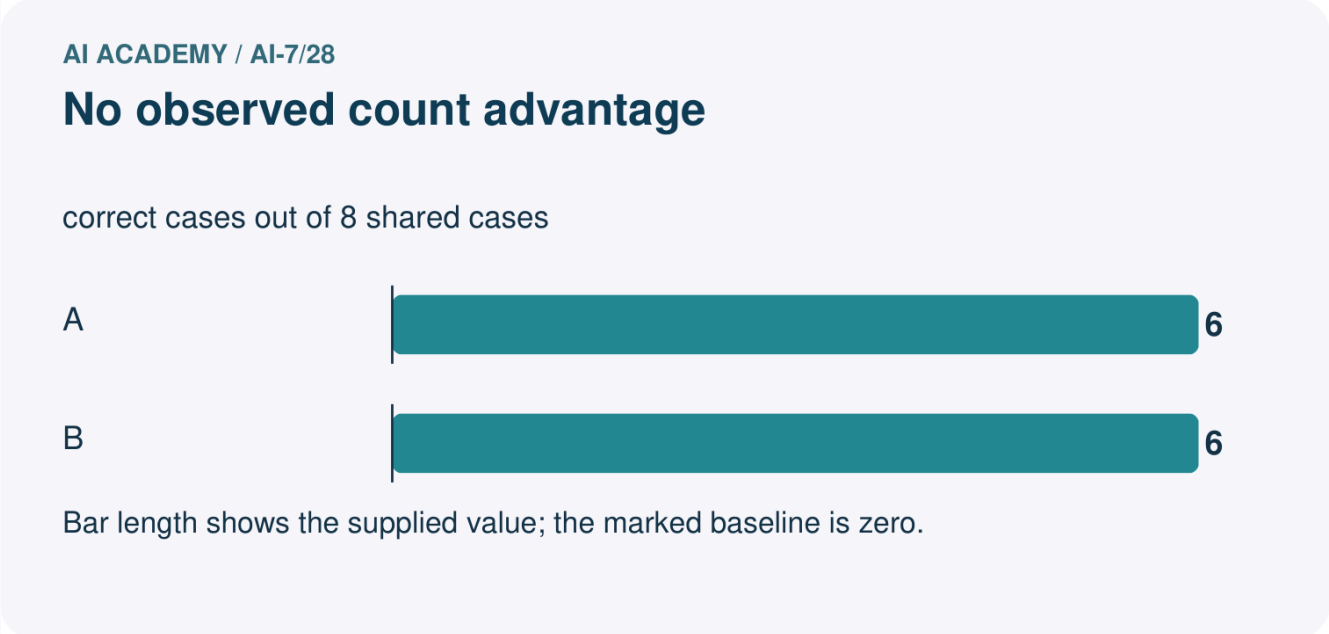
A pictorial worked case: Negative results and claim revision

A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

The facts we will use

Supplied facts

A and B each score 6/8. Before running, the team predicted B would score higher.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/28

Negative results and claim revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recall the prediction

The claim was a higher correct count for B.



2. State the finding

Both versions score 75%; no observed count advantage appears here.



3. Revise cautiously

Report the unsupported prediction and a justified next test.

CHANGE ONE THING

B wins on one new task after the team chooses it because A fails.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Recall the prediction	The claim was a higher correct count for B.
2. State the finding	Both versions score 75%; no observed count advantage appears here.
3. Revise cautiously	Report the unsupported prediction and a justified next test.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **negative result** can be informative when methods and limits are clear. A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

What this example does not establish

Keep the stated limit

A tie on eight cases is not an **equivalence** test; uncertainty and practical thresholds remain unspecified.

Keep the stated limit

Read the labelled picture: Does a 6/8 tie support the **prediction** that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove **equivalence**.

Keep the stated limit

Change one thing: B wins on one new task after the team chooses it because A fails. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Negative results and claim revision claim and explain a repair using the supplied evidence.

Failure to meet a criterion or reject a null is not proof of exact **equivalence**.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Failure to meet a criterion or reject a null is not proof of exact **equivalence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Negative results and claim revision with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Negative result**s and **claim revision**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Negative result**s and **claim revision**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does a 6/8 tie support the **prediction** that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove **equivalence**.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Negative results and claim revision. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Hypothesis expects B support strictly above A. Both pass8/10 reserved cases. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	

Record	Your evidence
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Negative results and claim revision. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Failure to meet a criterion or reject a null is not proof of exact **equivalence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **negative result**s and **claim revision evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **negative result**s and **claim revision**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Negative results and claim revision case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Negative results and claim revision — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Hypothesis expects B support strictly above A. Both pass8/10 reserved cases.	Compare results with the original criterion and uncertainty assumptions. Preserve the failed hypothesis, discuss plausible mechanisms and label revised questions as new work.	Strict improvement is not observed; both80%. This does not prove population equality or universal uselessness.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Hypothesis predicts mean latency reduction at least2 seconds. Observed paired mean reduction .8. Report.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Hypothesis predicts mean latency reduction at least2 seconds. Observed paired mean reduction .8. Report.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Negative results and claim revision, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

By the end, I can	How I will show it
Explain the mechanism and terminology of negative results and claim revision	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Negative results and claim revision without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **negative result**s and **claim revision**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report the unsupported **prediction** and a justified next test.

1. Revise cautiously

2. Recall the **prediction**

3. State the finding

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Revise cautiously Report the unsupported **prediction** and a justified next test. A **negative result** can be informative when methods and limits are clear.

A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: B wins on one new task after the team chooses it because A fails. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Treat that selected demonstration as exploratory, not a replacement for the original result.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Treat that selected demonstration as exploratory, not a replacement for the original result. Case selection now depends on outcomes.

Repair a missing building block

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

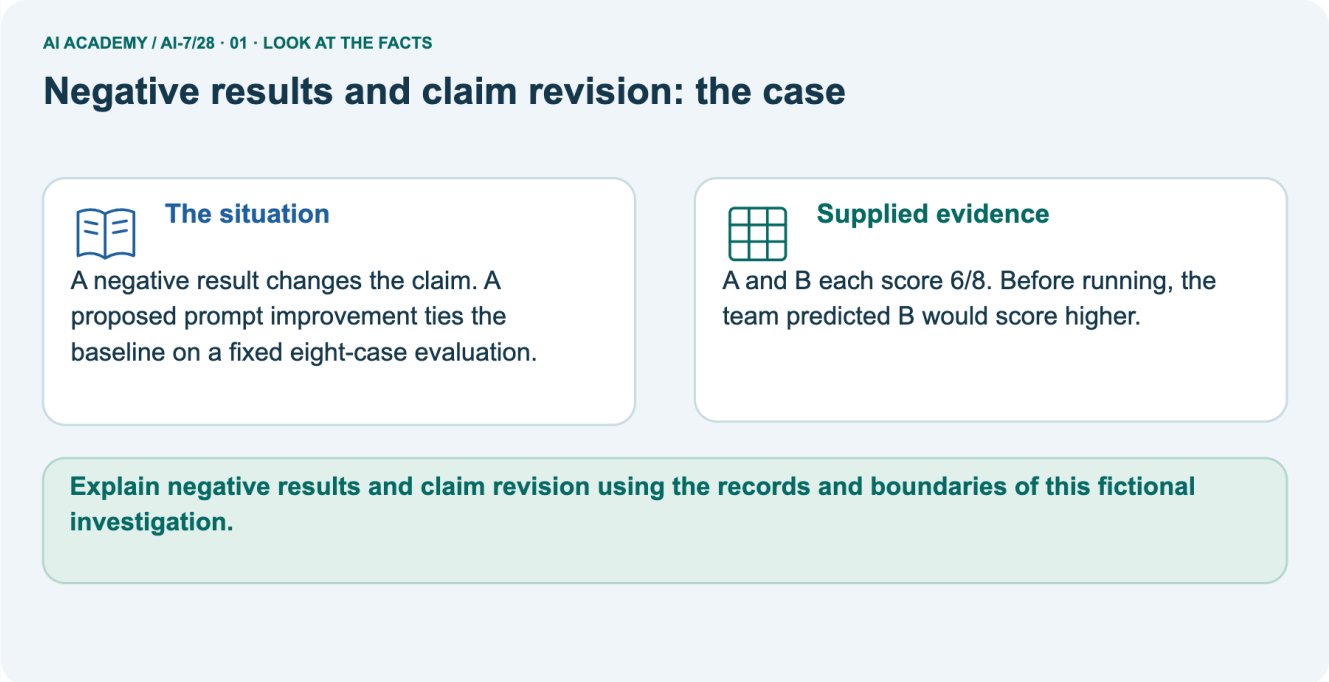
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Negative results and claim revision graphic using its labels, deciding evidence and relevant limitation.

01. Negative results and claim revision: the case

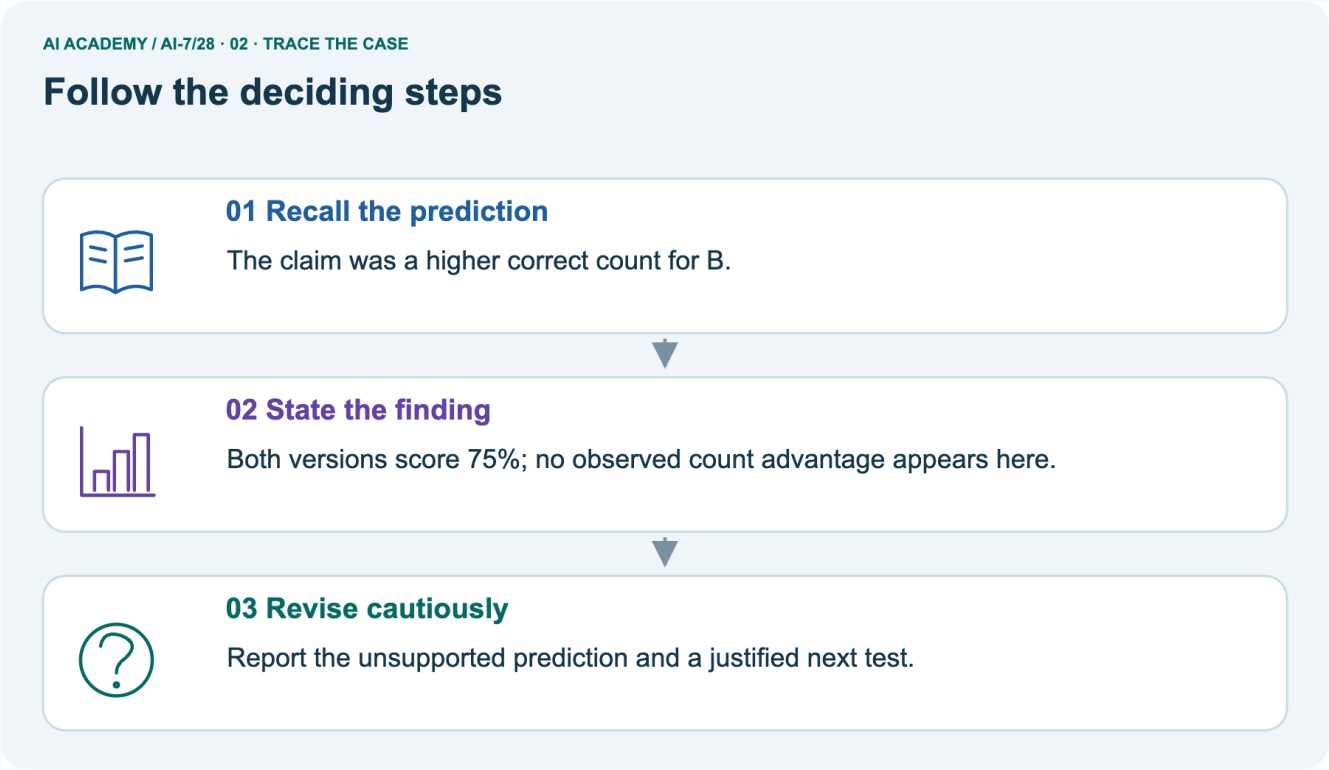


Follow the labelled picture with the matching explanation. Open the vector diagram

A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation. Supplied facts: A and B each score 6/8. Before running, the team predicted B would score higher.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

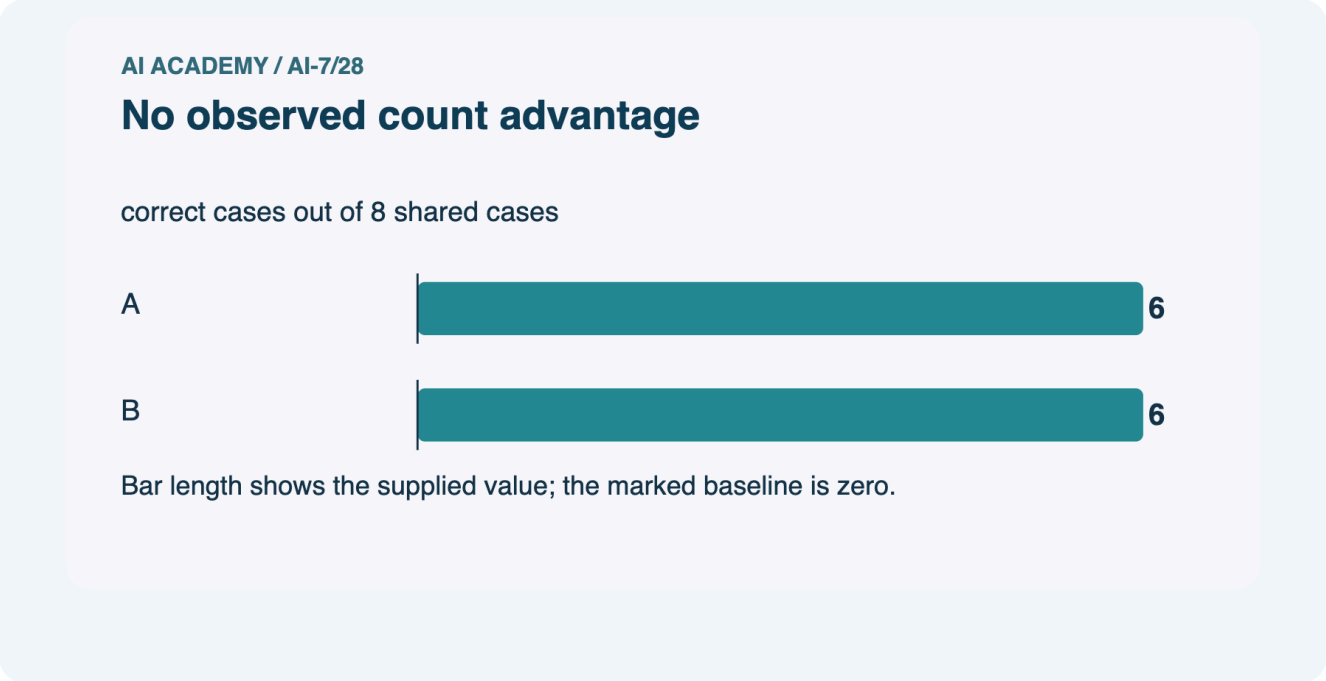


Follow the labelled picture with the matching explanation. Open the vector diagram

Recall the **prediction**: The claim was a higher correct count for B. → State the finding: Both versions score 75%; no observed count advantage appears here. → Revise cautiously: Report the unsupported **prediction** and a justified next test.

> Point to each numbered stage and explain which recorded **input** it uses.

03. No observed count advantage

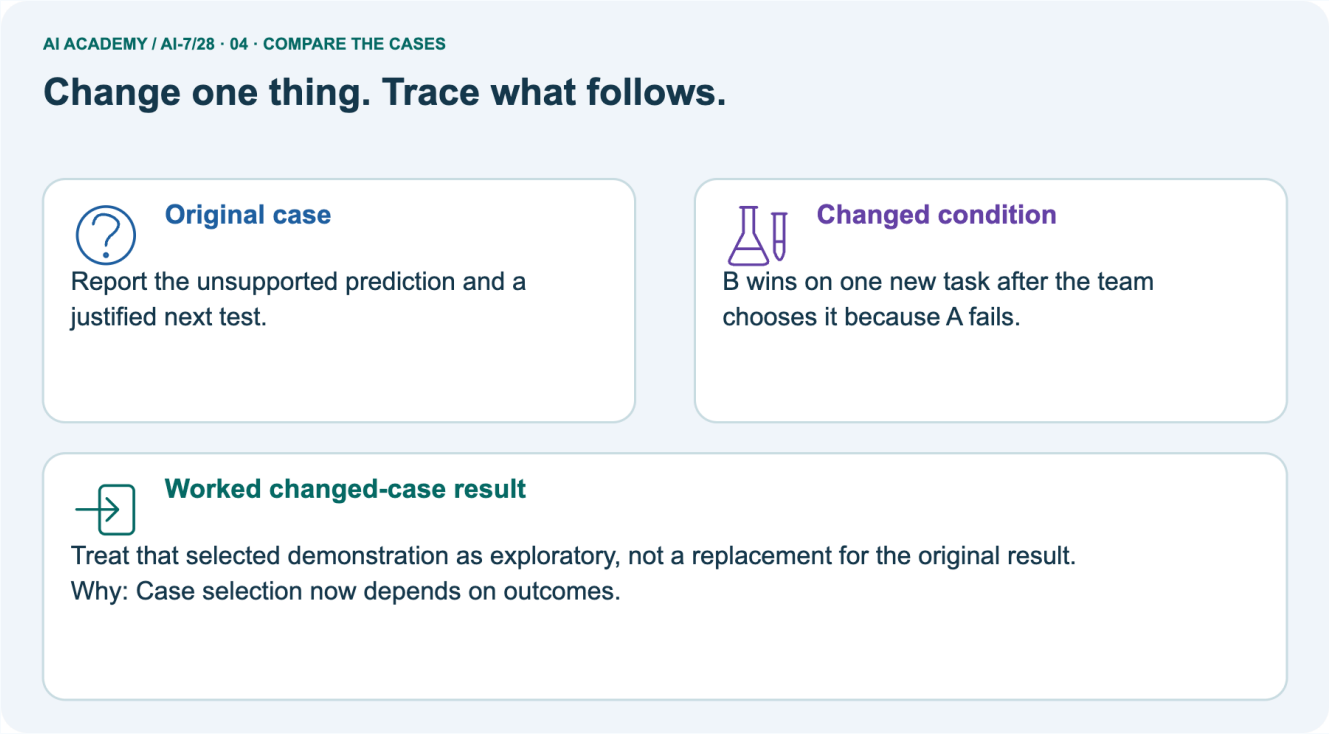


Follow the labelled picture with the matching explanation. Open the vector diagram

No observed count advantage. Units: correct cases out of 8 shared cases. A: 6. B: 6.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



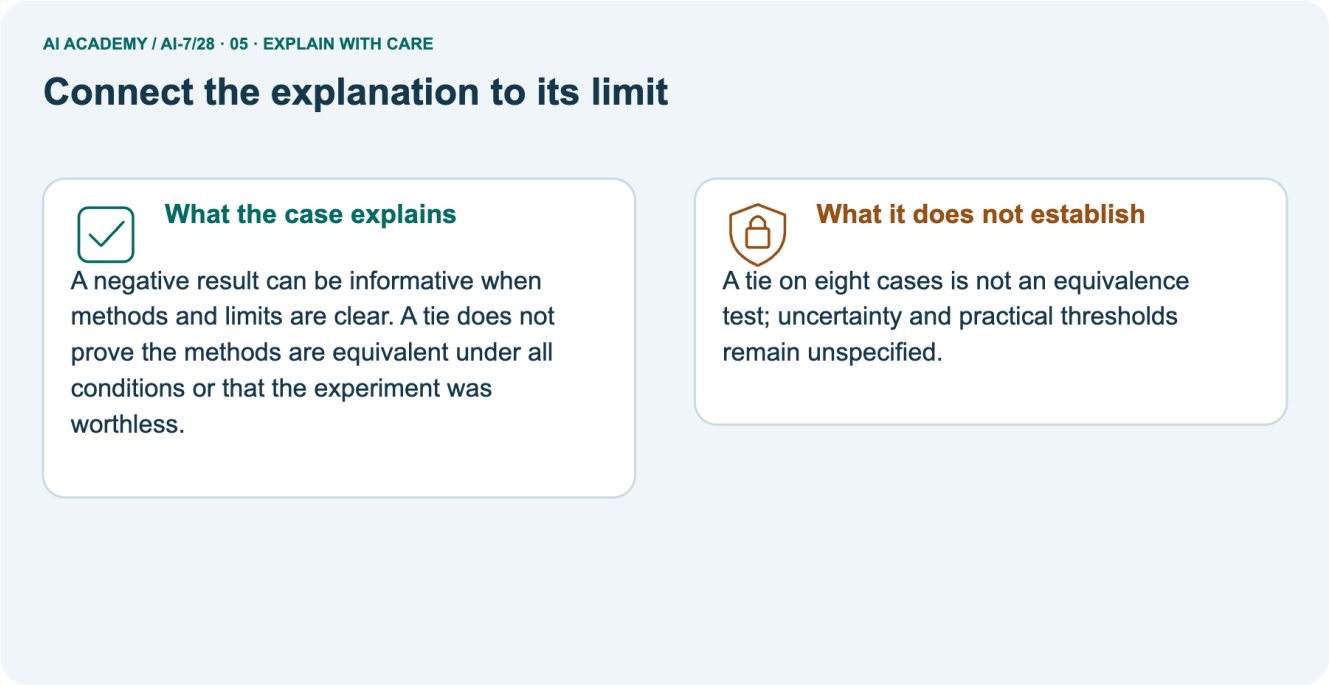
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report the unsupported **prediction** and a justified next test. Changed condition: B wins on one new task after the team chooses it because A fails.

Expected result: Treat that selected demonstration as exploratory, not a replacement for the original result. Reason: Case selection now depends on outcomes.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



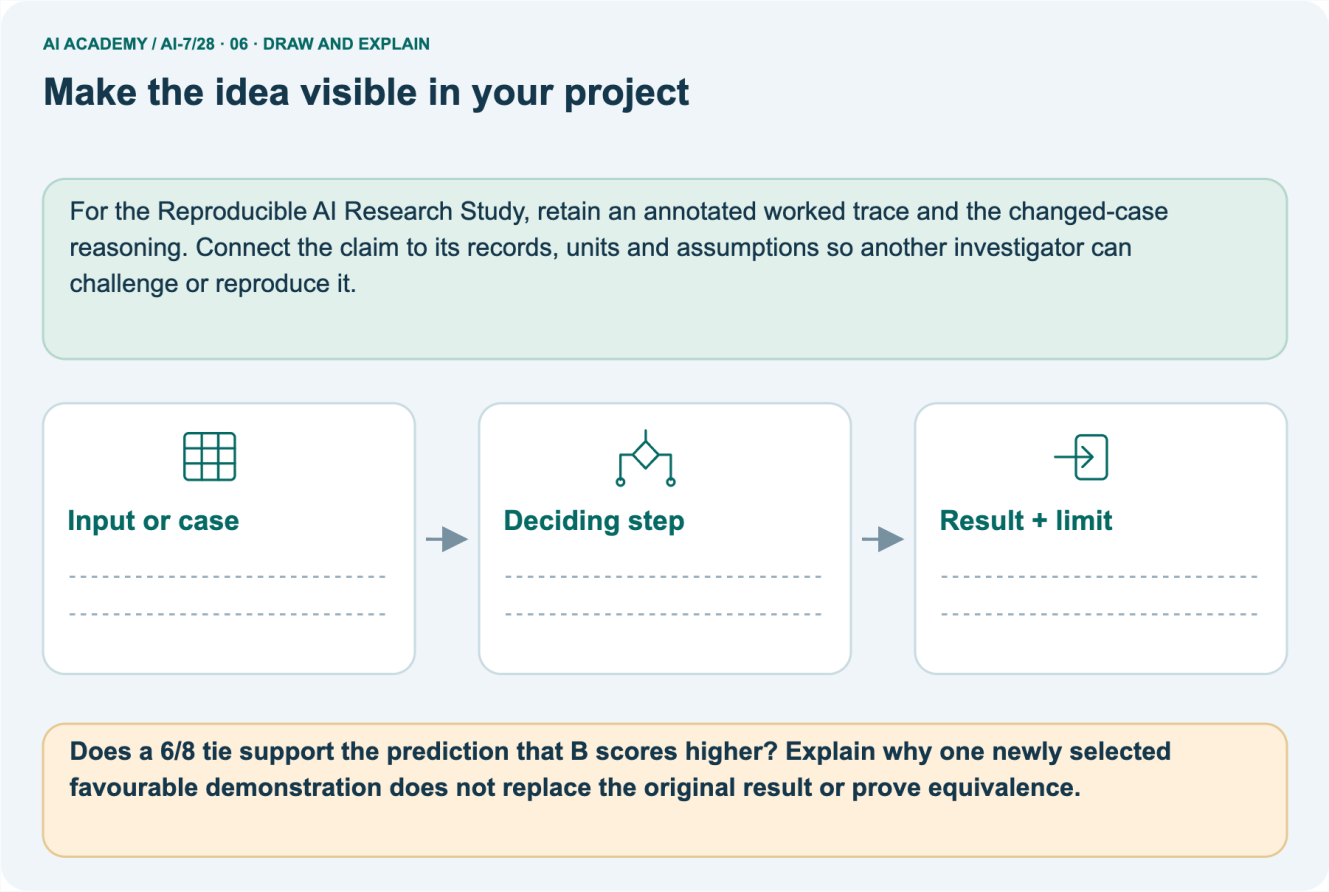
Follow the labelled picture with the matching explanation. Open the vector diagram

A **negative result** can be informative when methods and limits are clear. A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

Limit: A tie on eight cases is not an **equivalence** test; uncertainty and practical thresholds remain unspecified.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove **equivalence**.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.